

# A Snapshot-Conditioned CVAE Limit Order Book Simulator: Evidence from Canadian Equities

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## Abstract

Testing algorithmic trading strategies in production carries substantial risk of market disruption, yet passive historical replay cannot capture the market’s reaction to an agent’s actions. We propose a reactive limit order book (LOB) simulator based on a Conditional Variational Autoencoder (CVAE) coupled with an LSTM module that conditions generation on the ten most recent book snapshots. The model operates directly on state transitions signed, normalized depth vectors at three bid and ask levels rather than on heterogeneous order-by-order events, yielding a compact and homogeneous learning target. We train and evaluate on Toronto Stock Exchange (TSX) equity data, a market underrepresented in the simulation literature. The simulator reproduces key stylized facts: marginal queue distributions, the characteristic LOB smile in average quantities, level-to-level correlations, and realistic price paths—while exhibiting reactive price impact for both liquidity taking and provision, with the concave size-impact relationship qualitatively consistent with the literature. Benchmarked against a Conditional Wasserstein GAN with gradient penalty trained on the same data, the CVAE achieves superior queue-level fidelity and inter-level correlation structure with a simpler, single-network ELBO objective that requires neither gradient-penalty tuning nor critic-iteration scheduling, at the cost of a higher marginal KL divergence on the more liquid of the two stocks studied. To the best of our knowledge, this snapshot-conditioned CVAE+LSTM simulator is the first of its kind applied to Canadian equities, offering reactive synthetic environments for execution and market-making research.

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## 1. Introduction

A realistic simulator should understand and reproduce the stylized facts of the limit order book (LOB) so that trading strategies can be tested in a reactive environment rather than in passive market replay, which does not respond to an agent’s actions (Coletta et al., 2023; Jain et al., 2024). Such reactivity reduces reliance on risky production testing, avoiding flash-crash-like behavior, missed opportunities, or unintended market impact—and supports scenario exploration even under evolving regulations. As emphasized in authoritative surveys, there remains a persistent gap between empirical stylized facts and many existing models (Gould et al., 2013; Jain et al., 2024).

From the early developments in econophysics and order-driven models (e.g., Mike and Farmer (2008) and even before 2000 Bak et al. (1997)) aimed at reproducing stylized facts, research evolved toward tractable queueing approaches—most notably Markovian LOB models where market, limit, and cancel flows are represented through point processes, enabling analytic characterization of durations, autocorrelations, and diffusion limits (Cont et al., 2010; Cont and de Larrard, 2013). A key refinement, the queue-reactive model, conditions event intensities on the book state and has become a practical baseline for simulation and execution-cost analysis (Huang et al., 2015; Bodor and Carlier, 2025), though independent arrivals and simplified queue dynamics limit representation of clustered activity and herding.

To address these shortcomings, Hawkes-based models introduced self- and cross-excitation to capture clustering in order flow; later high-dimensional and marked versions incorporated volume, side, and level information, while recent work couples Neural Hawkes or LSTM embeddings to simulate event-driven LOB dynamics within deep-RL market-making frameworks (Abergel and Jedidi, 2015; Lu and Abergel, 2018; Chung et al., 2024;

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Lalor and Swishchuk, 2025). Yet despite these advances, Hawkes models often struggle to represent strategic interaction among heterogeneous agents.

Agent-based modeling (ABM) emerged partly in response, offering interpretability and explicit behavioral mechanisms. Byrd et al. introduced ABIDES, a high-fidelity NASDAQ-style simulator (ITCH/OUCH) that enables reactive experiments at scale, though models like Preis et al. (2006) had already differentiated liquidity providers from takers. More recent hybrid approaches combine neural point processes or diffusion models with ABIDES-like infrastructures to preserve stylized facts while enabling strategic agent interaction (Shi and Cartledge, 2024), though such approaches may require heavy calibration and often lack the statistical efficiency of deep generative models.

Deep learning and generative modeling further bridge realism and controllability: CNN/LSTM architectures such as DeepLOB capture spatial-temporal LOB structure, while RNN-based generative models reproduce transitions (type, price level, size, delay). Conditional GANs and transformers enhance reactivity, robustness, and short-horizon forecasting (Zhang et al., 2019; Hultin et al., 2023; Coletta et al., 2023; Xiao et al.). These developments complement queueing and Hawkes foundations and have become common baselines, supported by progress in scalable computation.

Much of the literature models events (limit, market, cancellation), sampling inter-event times and attributes order-by-order, yielding heterogeneous outputs that mix categorical order types, discrete prices, and continuous quantities/interarrival times. An alternative is to generate the next LOB snapshot directly, structuring inputs and outputs around state transitions between book states. This homogenizes the learning target, simplifies evaluation against stylized facts, and integrates naturally with execution modules.

We propose a snapshot-conditioned generative simulator of the limit order book. Rather than modelling order-by-order events, we represent the book state as a signed, normalised depth vector at the top  $k$  bid and ask levels,  $\mathbf{x}_t = (B_k, \dots, B_1, A_1, \dots, A_k) \in \mathbb{R}^{2k}$ , and learn a generative model of the state transition  $\mathbf{x}_t \rightarrow \mathbf{x}_{t+\Delta}$ ; this homogenises the learning target, removes the need to handle heterogeneous output types, and integrates naturally with execution and market-making modules. On top of this representation we couple a Conditional VAE (Kingma and Welling, 2014; Sohn et al., 2015) with a single-layer LSTM (Hochreiter and Schmidhuber, 1997) that summarises the  $h$  most recent snapshots into a history embedding, from which the decoder samples the next snapshot. Training maximises a  $\beta$ -weighted ELBO with a per-sample reconstruction weight that upweights the comparatively rare price-change events, and an ablation over the conditioning horizon  $h \in \{1, 2, 4, 10, 20\}$  confirms the importance of temporal context.

We evaluate the simulator on two Toronto Stock Exchange equities, BMO and Telus, chosen to represent contrasting liquidity regimes. It reproduces four key stylised facts—marginal queue distributions, the average-quantity smile, level-to-level correlations, and realistic price paths—and exhibits reactive price impact findings for both liquidity taking and provision. We further benchmark it against a Conditional WGAN-GP trained on the same data and given the same conditioning input, and characterise the resulting trade-off explicitly: the adversarial model attains a lower marginal KL divergence on BMO (0.054 vs. 0.143), whereas the CVAE attains a lower queue error (BMO: 34.1 vs. 55.3; Telus: 52.8 vs. 162.5) and a better correlation fidelity (BMO: 0.34 vs. 0.44; Telus: 0.25 vs. 0.71 stock units) under an objective that requires neither gradient-penalty tuning nor critic-iteration scheduling.

The novelty we claim rests on the gap identified above and is twofold. First, the deep generative simulators closest to ours—RNN- and GAN-based conditional generators (Hultin et al., 2023; Coletta et al., 2023; Cont et al., 2023)—generate order-by-order event streams, with the heterogeneous output space this entails; modelling the state transition itself on a fixed price grid, conditioned on an LSTM summary of the recent snapshot history, is to the best of our knowledge untried in this literature. Second, that literature is calibrated on US and European venues, leaving Canadian equities unexamined despite a large-tick regime and thinner books that make the snapshot representation particularly natural.

The remainder of the paper is organised as follows. Section 2 introduces the limit order book, the centred snapshot representation and its state transitions, and the CVAE+LSTM architecture. Section 3 presents the weighted  $\beta$ -ELBO training objective. Section 4 describes the TSX data and the model configuration, and Section 5 reports the stylised facts together with the reactivity experiments. Section 6 benchmarks the simulator against the Conditional WGAN-GP, Section 7 quantifies the contribution of the conditioning horizon, and Section 8 concludes and outlines directions for future work.

## 2. Model

### 2.1. Limit Order Book (LOB) Overview

A limit order book is a centralised record of outstanding buy and sell interest in a financial instrument at discrete price levels. Buy orders, which express a willingness to acquire the instrument, are stored on the bid side, while sell orders, expressing a willingness to dispose of it, are stored on the ask (or offer) side. The mechanism governing execution follows a price–time priority rule: among all pending orders, the highest bid and lowest ask receive priority, and ties at the same price are broken according to the time of arrival in a First-In-First-Out fashion. Table 1, inspired from the TSX Market Making Program Guide (?), illustrates a typical book state with four visible levels on each side.

Table 1: Illustration of LOB state

| Bid       |            |              | Ask       |           |              |
|-----------|------------|--------------|-----------|-----------|--------------|
| Level     | Qty        | Price        | Level     | Qty       | Price        |
| <b>L1</b> | <b>300</b> | <b>10.19</b> | <b>L1</b> | <b>50</b> | <b>10.22</b> |
| L2        | 200        | 10.18        | L2        | 500       | 10.23        |
| L3        | 400        | 10.17        | L3        | 300       | 10.24        |
| L4        | 500        | 10.16        | L4        | 200       | 10.25        |

In this example, the bid–ask spread—defined as the difference between the best ask and the best bid—is  $10.22 - 10.19 = 0.03$ , and the mid-price, computed as the average of the two best quotes, is 10.205. The evolution of the book is driven by three fundamental event types:

- **Market order:** a market order requests immediate execution at the best available opposite quote. It guarantees execution but offers no control over the price obtained. When the order size exceeds the quantity available at the best price, the residual fills against deeper levels, consuming liquidity and potentially shifting the best quote—a phenomenon known as market impact.
- **Limit order:** a limit order specifies both a price and a quantity at which the participant is willing to trade. If the specified price crosses the opposite best quote, the order triggers an immediate transaction; if it falls inside the spread, it improves the best displayed price; otherwise, it rests passively in the book at the designated level, awaiting a future counterparty.
- **Cancellation:** a cancellation removes a previously submitted limit order from the book. Participants frequently cancel and re-submit orders to reposition themselves at more favourable price levels or to manage inventory risk.

When an incoming order consumes liquidity across one or more price levels, the resulting shift in the mid-price is termed market impact; its magnitude depends on the order size relative to the available depth at each level (Farmer et al., 2005). Prices, however, do not move continuously: they evolve on a discrete grid whose minimum increment—the tick size—is set by market regulation. In Canadian and US equity markets, the tick size is  $\delta = 0.01$ .

Building on the standard event-based representation of electronic markets, we model each limit order as a triplet  $x = (t, p, q)$ , where  $t \in \mathbb{R}$  denotes the order submission time,  $p \in \delta\mathbb{N}$  denotes the limit price expressed on the discrete price grid induced by the tick size  $\delta > 0$ , and  $q \in \mathbb{Z} \setminus \{0\}$  denotes the signed order quantity. By convention,  $q > 0$  corresponds to an ask (sell) order, while  $q < 0$  corresponds to a bid (buy) order.

Let  $\mathbf{L}$  denote the limit order book (LOB), understood as the collection of all active limit orders that have been submitted but not yet fully executed or cancelled. For any  $t \in \mathbb{R}^+$ , we denote by  $\mathbf{L}(t)$  the complete state of the LOB at time  $t$ .

#### 2.1.1. Queue Size and Best Prices

For any price level  $p \in \delta\mathbb{N}$  and time  $t$ , the aggregate queue size at price  $p$  is defined as

$$Q_p(t) = \sum_{x \in \mathbf{L}_p(t)} q_x,$$

where

$$\mathbf{L}_p(t) = \{x \in \mathbf{L}(t) : p_x = p\}$$

denotes the set of all outstanding orders posted at price  $p$  at time  $t$ .

Positive queue sizes correspond to sell-side liquidity, while negative queue sizes correspond to buy-side liquidity. The prevailing best ask and best bid prices are defined respectively as

$$p_a(t) = \inf\{p \in \delta\mathbb{N} : Q_p(t) > 0\}, \quad p_b(t) = \sup\{p \in \delta\mathbb{N} : Q_p(t) < 0\}.$$

The mid-price is then given by

$$p_{\text{mid}}(t) = \frac{p_a(t) + p_b(t)}{2}.$$

By construction, the bid–ask spread is strictly positive, so that  $p_a(t) > p_b(t)$  for all  $t$  at which the market is well defined.

### 2.1.2. Centered Limit Order Book Snapshot

To obtain a finite-dimensional representation of the LOB, we consider a symmetric price grid of length  $2k$  whose lowest level  $p$  is the best bid minus  $(k - 1)$  ticks—equivalently, the deepest bid price in the window. Under the assumption of a one-tick spread, the grid is centred on the mid-price  $p + (k - \frac{1}{2})\delta$ . Specifically, the grid consists of the price levels

$$p, p + \delta, p + 2\delta, \dots, p + (2k - 1)\delta.$$

The centered LOB snapshot at time  $t$  is defined as the vector

$$X_t = (Q_p(t), \dots, Q_{p+(k-1)\delta}(t), Q_{p+k\delta}(t), \dots, Q_{p+(2k-1)\delta}(t)).$$

The grid is chosen such that the first  $k$  components correspond to bid-side queues and the remaining  $k$  components correspond to ask-side queues. Consequently, we impose the sign constraints

$$Q_{p+i\delta}(t) \leq 0 \quad \text{for all } i \in \{0, \dots, k - 1\},$$

$$Q_{p+i\delta}(t) \geq 0 \quad \text{for all } i \in \{k, \dots, 2k - 1\}.$$

Under this construction, the best bid queue corresponds to

$$Q_{p+(k-1)\delta}(t),$$

while the best ask queue corresponds to

$$Q_{p+k\delta}(t).$$

Throughout the analysis, and in line with common assumptions for large-tick assets, we assume that queues at the best bid and best ask prices are never empty. This ensures that the bid–ask spread remains well defined at all times. It is important to point out that it is assumed that no gap other than the tick size exists between the price levels.

### 2.1.3. State Transitions

Market participants' actions, namely submissions, cancellations, and executions of limit orders are discrete events in continuous time. Their cumulative effect over some appropriate time span  $\Delta t$  may be viewed as a continuous transition of the queue sizes from state  $X_t$  to  $X_{t+\Delta t}$ . Rather than simulating individual order arrivals, we model directly the distribution of the centered order book state at time  $t + \Delta t$ . Crucially, the price grid  $(p, p + \delta, \dots, p + (2k - 1)\delta)$  remains *fixed*; price movements are encoded through *sign changes* in the queue sizes. The future LOB snapshot on this fixed grid takes one of the following three forms.

*No price change..* If neither the best bid nor the best ask price changes during  $(t, t + \Delta t)$ , the reference price remains fixed. In this case, the state transition affects only the queue sizes, yielding:

$$X_{t+\Delta t} = \left( \underbrace{Q_p(t + \Delta t), \dots, Q_{p+(k-1)\delta}(t + \Delta t)}_{\leq 0 \rightarrow \text{bid}---}, \underbrace{Q_{p+k\delta}(t + \Delta t), \dots, Q_{p+(2k-1)\delta}(t + \Delta t)}_{\geq 0 \rightarrow \text{ask}+++} \right).$$

*Price decrease..* For a price decrease, one or more bid queues must turn positive, thereby becoming ask queues. The reference price shifts downward by  $\delta$ . The centered representation is updated accordingly, resulting in. For  $k = 3$  the resulting snapshot reads

$$X_{t+\Delta t} = \left( \underbrace{Q_p(t+\Delta t), Q_{p+\delta}(t+\Delta t)}_{\leq 0 \rightarrow \text{bid--}}, \underbrace{Q_{p+2\delta}(t+\Delta t), Q_{p+3\delta}(t+\Delta t), Q_{p+4\delta}(t+\Delta t), Q_{p+5\delta}(t+\Delta t)}_{\geq 0 \rightarrow \text{ask++++}} \right).$$

Similarly, a price decrease of  $m \leq k$  ticks is induced by  $m$  former bid quantities flipping their sign.

*Price increase..* Symmetrically, for a price increase one or more ask queues must turn negative, thereby becoming bid queues. A one-tick price increase requires  $Q_{p+k\delta}(t+\Delta t) < 0$ . For  $k = 3$  the snapshot becomes

$$X_{t+\Delta t} = \left( \underbrace{Q_p(t+\Delta t), Q_{p+\delta}(t+\Delta t), Q_{p+2\delta}(t+\Delta t), Q_{p+3\delta}(t+\Delta t)}_{\leq 0 \rightarrow \text{bid----}}, \underbrace{Q_{p+4\delta}(t+\Delta t), Q_{p+5\delta}(t+\Delta t)}_{\geq 0 \rightarrow \text{ask++}} \right).$$

A price increase of  $m$  ticks corresponds to  $m$  former ask quantities turning negative. Note that the subscripts are identical in all three cases because the grid does not shift; the three scenarios differ only in how the sign partition divides the vector.

The advantage of this fixed-grid representation is that price changes are inherently encoded in the sign structure of the snapshot vector, without introducing an additional discrete price-change variable. For large-tick stocks, price changes are discrete and generally span only a few ticks for an appropriate  $\Delta t$ , since typically only a small number of queues get depleted and refilled on the opposite side. At most  $k$  price changes during  $\Delta t$  can be represented, so  $k$  and  $\Delta t$  should be chosen to match the characteristics of each stock. For small-tick stocks, one may consider binning several levels into a single one. This yields a compact, homogeneous learning target that integrates naturally with the sequential LSTM architecture described in the following subsection.

*Grid re-centring during trajectory generation..* The training data is constructed from a wider book window (typically 4k levels) by selecting, at each timestamp, the  $k$  levels nearest the current mid-price. Consequently, a price-change snapshot—exhibiting a non-balanced sign partition such as  $(k-m)/(k+m)$ —is immediately followed in the dataset by a snapshot re-centred on the new mid-price, restoring the canonical  $k/k$  partition. Since the conditioning sequences seen during training contain these sign-flip-then-rebalance transitions, the model learns to produce appropriately centred outputs after receiving a price-change input; no explicit post-processing is required during autoregressive generation. The representation can accommodate at most  $k$  consecutive same-direction moves within a single conditioning window; for the stocks and time scale studied ( $\Delta t = 1$  s), this limit is never reached in practice.

#### 2.1.4. Distribution

The next LOB state is a random variable whose distribution is conditional on the recent state history  $X_t, X_{t-\Delta t}, \dots, X_{t-h\Delta t}$ , denoted  $S_t = (X_{t-h\Delta t}, \dots, X_{t-\Delta t}, X_t)$ :

$$X_{t+\Delta t} | S_t \sim \mathbf{Pr}(x_{t+\Delta t} | S_t).$$

We model this conditional distribution using a Conditional Variational AutoEncoder (CVAE) (Kingma and Welling, 2014; Sohn et al., 2015), for three complementary reasons. First, the CVAE is trained by maximising a single scalar objective derived from the evidence lower bound (ELBO), avoiding the min-max saddle-point optimisation of GANs and the associated training instabilities such as mode collapse and sensitivity to critic-iteration scheduling. . Second, the stochastic latent variable  $\mathbf{z}$  allows the model to represent the inherent multimodality of LOB transitions: the same history  $S_t$  can plausibly evolve into qualitatively distinct next states depending on unobserved order flow, and sampling different values of  $\mathbf{z}$  from the learned prior recovers this diversity without requiring explicit mixture components. Third, the conditional encoder-decoder structure of the CVAE integrates naturally with the LSTM-derived history embedding  $\mathbf{h}_t = \text{LSTM}(S_t)$ : the embedding is passed as the conditioning variable  $\mathbf{y}$ , so the single-network architecture requires neither a critic nor alternating training schedules. The standard VAE learns a probabilistic mapping from a data space  $\mathbf{x}$  to a latent space  $\mathbf{z}$ , where both the encoder and decoder are parameterized by neural networks. The goal of the VAE is to maximize the marginal likelihood  $p(\mathbf{x})$ , which can be expressed as

$$p(\mathbf{x}) = \int p(\mathbf{x} | \mathbf{z}) p(\mathbf{z}) d\mathbf{z}.$$

However, directly computing this marginal is intractable, so we use the variational approximation

$$q(\mathbf{z} \mid \mathbf{x}) \approx p(\mathbf{z} \mid \mathbf{x}).$$

Thus, the VAE objective function becomes the Evidence Lower Bound (ELBO):

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{q(\mathbf{z} \mid \mathbf{x})} [\log p(\mathbf{x} \mid \mathbf{z})] - D_{\text{KL}}(q(\mathbf{z} \mid \mathbf{x}) \parallel p(\mathbf{z})),$$

where  $q(\mathbf{z} \mid \mathbf{x})$  is the variational posterior,  $p(\mathbf{x} \mid \mathbf{z})$  is the likelihood (decoder),  $p(\mathbf{z})$  is the prior (typically a standard Gaussian), and  $D_{\text{KL}}$  is the Kullback–Leibler divergence between the variational posterior and the prior.

In a Conditional VAE, we extend the VAE framework by conditioning the model on some external information  $\mathbf{y}$ . This allows the model to generate data that is dependent on  $\mathbf{y}$ , for example, generating images given class labels. The encoder produces a distribution  $q(\mathbf{z} \mid \mathbf{x}, \mathbf{y})$ , while the decoder produces  $p(\mathbf{x} \mid \mathbf{z}, \mathbf{y})$ . Thus, the conditional VAE objective becomes

$$\mathcal{L}(\theta, \phi; \mathbf{x}, \mathbf{y}) = \mathbb{E}_{q(\mathbf{z} \mid \mathbf{x}, \mathbf{y})} [\log p(\mathbf{x} \mid \mathbf{z}, \mathbf{y})] - D_{\text{KL}}(q(\mathbf{z} \mid \mathbf{x}, \mathbf{y}) \parallel p(\mathbf{z} \mid \mathbf{y})),$$

where  $q(\mathbf{z} \mid \mathbf{x}, \mathbf{y})$  is the conditional posterior,  $p(\mathbf{x} \mid \mathbf{z}, \mathbf{y})$  is the conditional likelihood, and  $p(\mathbf{z} \mid \mathbf{y})$  is the prior conditioned on the external information. Both encoder and decoder are parameterised by neural networks.

## 2.2. LSTM

Volatility clustering refers to the long memory of absolute or squared mid-price returns and means that large changes in price tend to follow other large price changes. Bouchaud et al. (2002) explain this finding by the arrival of publicly available news which leads traders to perform similar actions on a short timescale. This feature of limit order books motivates the use of self- and cross-exciting processes such as the Hawkes model. One measure commonly used to characterise long memory is the Hurst exponent  $H$ : for the stationary increments  $X_t$  of a long-memory process with  $H \in (\frac{1}{2}, 1)$ , the autocorrelation decays as

$$\rho(\tau) = \text{Corr}(X_t, X_{t+\tau}) \sim c\tau^{2H-2}.$$

Long memory is also observed in limit order book snapshots, as illustrated by the 1-second bid queue partial autocorrelation at a 95 percent confidence level in Figure 1.

Queue sizes at individual price levels exhibit significant serial dependence: depth added or removed at one timestamp is strongly predictive of depth at subsequent timestamps, reflecting the persistence of limit-order placement and cancellation activity. Figure 1 displays the partial autocorrelation function (PACF) of the best-bid queue at one-second resolution, showing statistically significant coefficients well beyond lag 1. This establishes that a first-order Markov model—conditioning only on the most recent snapshot—cannot capture the serial structure of the queue process, and motivates conditioning on a window of  $h > 1$  past snapshots.

The choice of  $h$  is a practical compromise between representational capacity and computational cost, and a sensitivity analysis over  $h$  is provided in the ablation study (Section 7). Several ways of dealing with long memory exist, for instance ARMA-related models (ARIMA, SARIMA), exponential smoothing, and Kalman filtering (Kalman, 1960). On the deep learning front, one of the most renowned architectures for sequential data is the LSTM. The gated mechanism allows the network to selectively forget or retain information along the sequence. Other models include the gated recurrent unit (Chung et al., 2014), which is mathematically similar to the LSTM but with fewer parameters, the attention-based Transformer, which captures long-range dependencies through self-attention, and the Temporal Convolutional Network (Lea et al., 2017), which uses dilated convolutions. We adopt the LSTM (Hochreiter and Schmidhuber, 1997) for its proven ability to capture long-range dependencies in sequential financial data (Zhang et al., 2019). Through its gated cell mechanism, the LSTM processes the history sequence  $S_t$  and produces a fixed-length hidden state

$$\mathbf{h}_t = \text{LSTM}(S_t),$$

which serves as the history embedding passed to the CVAE encoder and decoder, in both cases by concatenation with their respective inputs—the snapshot  $\mathbf{x}_t$  on the encoder side, the sampled latent vector  $\mathbf{z}_t$  on the decoder side; we refer the reader to Hochreiter and Schmidhuber (1997) for the full gating equations.

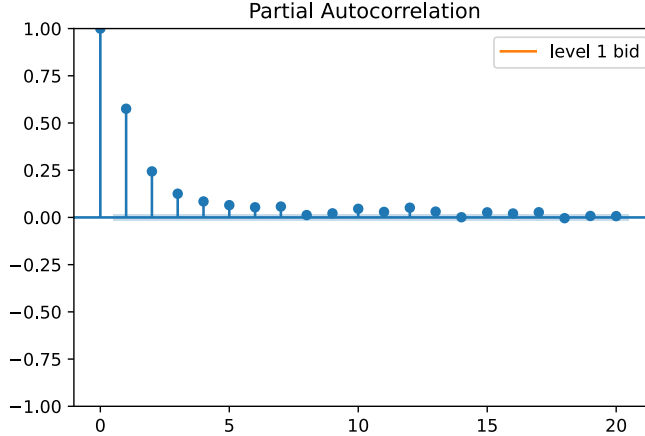


Figure 1: Partial Autocorrelation function - first level bid queue

### 3. Training Objective

Another compelling variation of the Variational Autoencoder (VAE) introduces a hyperparameter that explicitly balances the trade-off between the reconstruction loss and the Kullback-Leibler (KL) divergence term in the objective function. By adjusting this hyperparameter, the model gains greater flexibility during the learning process, allowing practitioners to fine-tune how strongly the latent space is regularized versus how accurately the input is reconstructed Higgins et al. 2017.

With this modification, the standard VAE loss function is reformulated as

$$\mathcal{L} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})] - \beta \cdot D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})),$$

where  $\beta$  is the newly introduced hyperparameter controlling the relative importance of the KL divergence term.

The data, after normalization, tends to be centered around the zero axis, with the majority of the mass concentrated on the positive side for *ask* queues and the negative side for *bid* queues. Smaller clusters of mass correspond to price changes occurring between consecutive snapshots. These events, while relatively rare, are significant and their magnitude depends on the rate of price fluctuations per timestep.

To better capture and emphasize these rare but important events during training, the VAE objective can be modified to penalize reconstruction errors more heavily for inputs associated with price changes. The adjusted loss function is expressed as

$$\mathcal{L} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [w(\mathbf{x}) \cdot \log p_\theta(\mathbf{x}|\mathbf{z})] - \beta \cdot D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})), \quad (1)$$

where  $w(\mathbf{x})$  is a sample-dependent weighting function, set to  $w_c > 1$  for snapshots corresponding to price changes and to 1 otherwise, so that reconstruction errors on these rare events are penalised more heavily;  $\beta$  controls the strength of the KL-divergence regularization, balancing reconstruction fidelity against latent space regularization;  $q_\phi(\mathbf{z}|\mathbf{x})$  denotes the approximate posterior; and  $p_\theta(\mathbf{x}|\mathbf{z})$  is the likelihood (decoder) model.

By assigning a higher weight to the rare price-changing samples, the VAE is encouraged to model these important transitions with greater accuracy. This methodology is inspired from Chawla et al. (2002).

## 4. Numerical experiments

### 4.1. Data

We use Canadian equity market data centered around the year 2016. This choice is motivated by the limited availability of limit order book (LOB) data, which is generally difficult to access and not readily obtainable through public online sources. Despite that, the dataset remains comprehensive and sufficiently rich in microstructure information, providing meaningful insights into order flow dynamics and market behavior. The LOB is preprocessed and represented as a sequence of fixed-lag 1-second snapshots, each snapshot encoded as a signed, normalized depth vector at the top three bid and ask levels,

$$\mathbf{x}_t = (B3_t, B2_t, B1_t, A1_t, A2_t, A3_t) \in \mathbb{R}^6,$$

with bids negative and asks positive as presented earlier. The snapshot methodology (as opposed to order-by-order simulation) treats order characteristics and inter-arrival times as *implicit* in the state transition  $\mathbf{x}_t \rightarrow \mathbf{x}_{t+\Delta}$ , significantly reducing dimensionality and yielding a homogeneous learning target focused on book-state dynamics.

Following the snapshot construction described in the simulator deck, each trading day is reduced to an effective window by clipping the first and last 30 minutes of trading, yielding approximately 5.5 hours per day. This choice avoids mixing distinct microstructure regimes typically observed near the open and close. Excluding the opening and closing half-hours is standard practice in the empirical microstructure literature: the call-auction mechanisms at open and close, elevated adverse selection immediately after the open, and end-of-day inventory rebalancing at the close generate order-flow and volatility patterns that are qualitatively distinct from the continuous intraday session (Gould et al., 2013; Cont et al., 2010; Huang et al., 2015). The snapshot-based approach highlights a limitation at sub-second resolutions: price changes tend to be infrequent relative to long stretches of unchanged states, making them difficult for a model to learn reliably. This motivates the use of second-scale (or coarser) sampling and/or longer temporal conditioning. In practice, depending on the stock, higher-frequency price changes may not occur at all, which can require propagating (or “dragging”) the last observed snapshot forward until a genuine state change is observed. Moreover, this approach implicitly assumes no gaps between price levels and a perfectly tight bid–ask spread of one tick. In real markets, these assumptions are often violated, as gaps and wider spreads do occur; addressing these discrepancies and their implications is deferred to a later discussion in the document. The final dataset is comprised of 20 days of 1-second clipped data and uses a 80-20 training validation scheme. Price paths are generated by seeding the conditioning window with a randomly drawn snapshot sequence from the dataset, then sampling autoregressively from the decoder.

#### 4.2. Models

Rather than conditioning only on the most recent snapshot (Markov), we condition on the last  $h = 10$  snapshots:

$$\mathbf{X}_t^{(h)} = (\mathbf{x}_{t-(h-1)\Delta}, \dots, \mathbf{x}_t), \quad h = 10,$$

and summarize this history through an LSTM hidden state

$$\mathbf{h}_t = \text{LSTM}(\mathbf{X}_t^{(h)}),$$

using a single layer with hidden size 128.

*Conditional VAE (latent variable and decoder MLP).* Given the history embedding  $\mathbf{h}_t$ , the CVAE models the next snapshot as

$$p_\theta(\mathbf{x}_{t+\Delta} \mid \mathbf{h}_t) = \int p_\theta(\mathbf{x}_{t+\Delta} \mid \mathbf{h}_t, \mathbf{z}_t) p(\mathbf{z}_t) d\mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where the latent dimension is explored over 8–32. The decoder is implemented as a lightweight MLP head on top of the conditioning and latent variables; we explore MLP hidden widths in the range 32–64. The retained configuration and the full training setup are reported in Table 2.

### 5. Results

#### 5.1. Marginal distribution

The primary objective of the model is to learn the marginal distribution of the observed data. Two elements of the training objective introduced in Section 3 bear directly on this task: the  $\beta$  weighting, whose low value relaxes the KL regularisation and thereby favours reconstruction fidelity and the preservation of fine-grained patterns, and the per-sample weight  $w_c$ , which upweights the rare price-change events populating the tails of the distribution—regions that are disproportionately important because they carry the price movements the simulator must reproduce.

It is also important to note the structural characteristics of the data itself. In many financial and market settings, quantities such as queue sizes are aggregated in discrete units, commonly referred to as “lots.” This aggregation inherently reduces the likelihood of observing very small queue sizes. As a result, the empirical distribution often exhibits a bimodal pattern, corresponding to positive and negative variations in these aggregated quantities. For modelling purposes, this bimodality can be effectively approximated by a unimodal

Table 2: Hyperparameters and training configuration

| Parameter                   | Value              |
|-----------------------------|--------------------|
| <i>Architecture</i>         |                    |
| LSTM layers                 | 1                  |
| LSTM hidden size            | 128                |
| Latent dimension            | 16                 |
| Decoder MLP hidden width    | 64                 |
| Conditioning horizon $h$    | 10                 |
| <i>Training</i>             |                    |
| Optimizer                   | Adam               |
| Learning rate               | $1 \times 10^{-3}$ |
| Batch size                  | 256                |
| Epochs                      | 100                |
| $\beta$ (KL weight)         | 0.1                |
| $w_c$ (price-change weight) | 2.0                |
| <i>CGAN Benchmark</i>       |                    |
| Generator hidden size       | 128                |
| Discriminator hidden size   | 128                |
| Critic iterations           | 5                  |
| Gradient penalty $\lambda$  | 10                 |
| Epochs                      | 300                |

distribution that encompasses both extremes, simplifying the representation while retaining the essential distributional properties. Tests were realized on both BMO and Telus stock but we will only present BMO in this document.

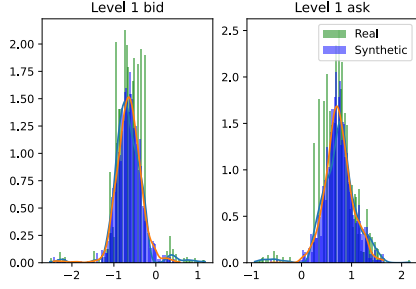


Figure 2: level 1 CVAE

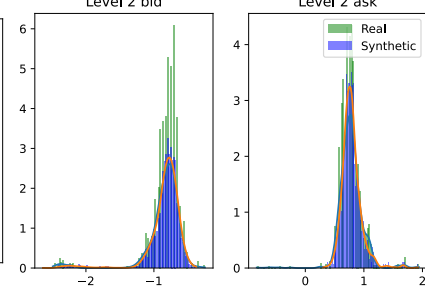


Figure 3: level 2 CVAE

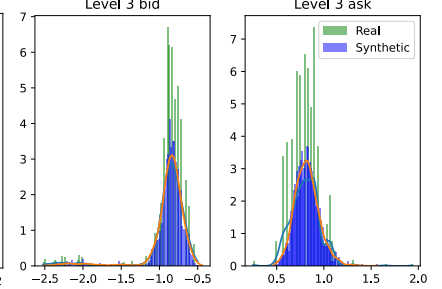


Figure 4: level 3 CVAE

As presented in Figures 2, 3, and 4, the model captures the overall shape and support of the marginal distributions at all three levels, including the bimodal bid–ask separation and the relative scale of deeper queues. However, the generated distributions are smoother than the empirical ones: the sharp peaks arising from lot-size quantisation (multiples of 100 shares) are not reproduced, and some tail mass—notably near  $-2.5$  at level 3 bid—is underrepresented. These discrepancies are consistent with the Gaussian decoder assumption, which favours unimodal outputs and inherently smooths discrete concentration points. The marginal KL divergence reported in Table 3 reflects this gap; improving tail fidelity through a more expressive decoder or a mixture-based prior is left to future work.

### 5.2. Average best quantities

One natural follow up from the marginal distribution performances is the ability to reproduce the order book quantities with high fidelity. This can be shown in Figure 5 where we can see that the smile shape typically presented by the LOB (see Abergel et al. 2016) is well learnt by the model. Differences by a magnitude of one round lot (100) can be noted on the first bid. This testify of the more volatile nature of the inner levels.

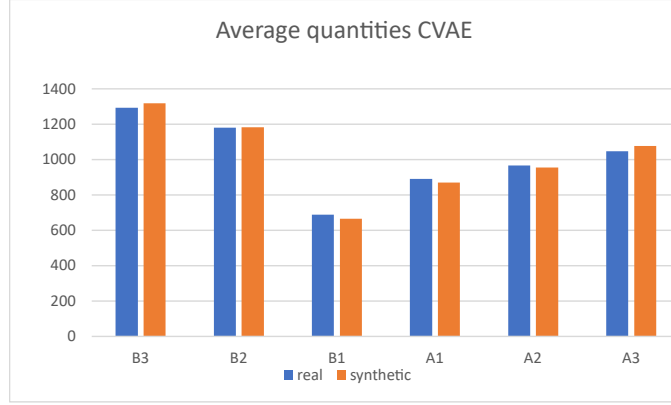


Figure 5: average best quantities CVAE

### 5.3. Level-to-level correlation

Level-to-level correlation plays a crucial role in capturing the full dynamics of the limit order book (LOB), as it reflects how changes at one price level influence—or are influenced by—changes at adjacent levels. This is particularly important in high-frequency trading environments, where the addition or removal of orders at one level can propagate through the book, affecting both liquidity distribution and price formation. This is the reason for the adoption of models such as multidimensional Hawkes processes, offering the ability to capture these relationships (Lu and Abergel, 2018).

One of the simplest measurements to quantify these relationships is the Pearson correlation coefficient. However, it is important to note that Pearson correlation measures only linear dependencies between variables. As a result, more complex, non-linear interactions—such as sudden liquidity shifts, strategic order placement, or hidden liquidity effects—may not be fully captured by this measure.

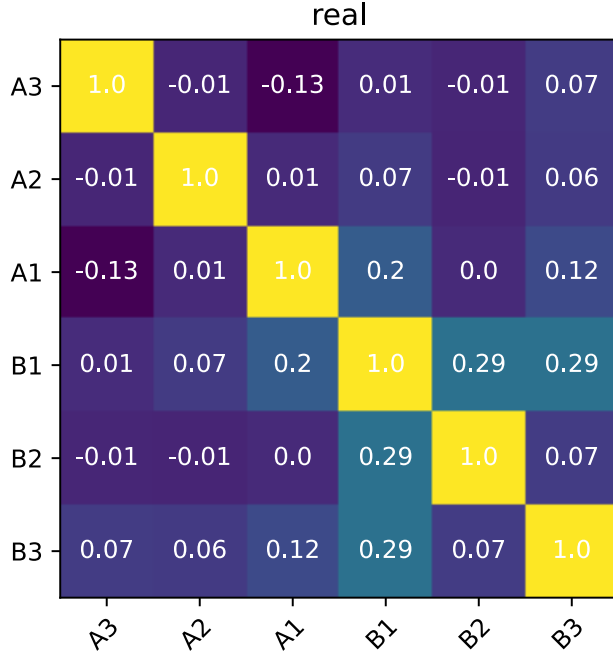


Figure 6: Correlations real CVAE

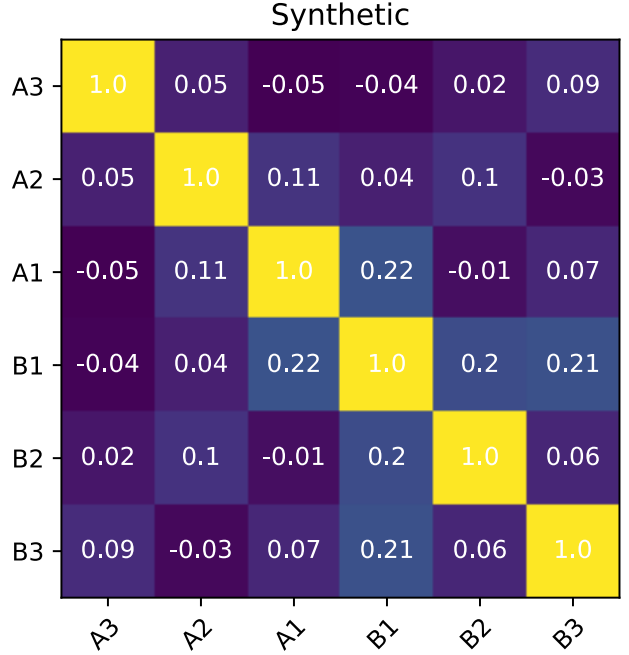


Figure 7: Correlation model CVAE

As we can see in Figures 6 and 7, the model-generated correlations closely match the real ones. From an architectural standpoint, a larger latent dimension gives the decoder more capacity to represent joint variation across levels; however, this also increases the risk of overfitting. The weighted reconstruction loss mitigates

this trade-off by steering gradient signal toward the rarer price-change snapshots. This involves coordinated sign flips across multiple levels, without requiring an excessively large latent space. We note that these are qualitative design considerations; a systematic ablation over the latent dimension is not performed in this work and is left to future investigation.

#### 5.4. Price paths

Price paths are a good indicator of the simulator’s usability as well as success in learning the LOB’s dynamic. The time series are obtained by sampling the decoder sequentially and translating the sign structure of the snapshot into a price change (measured in ticks, i.e.  $\delta = 1$ ) using

$$\Delta p_{t+\Delta t} = -\frac{1}{2} \sum_{i=1}^{2k} \text{sign}(X_{t+\Delta t, i}).$$

The factor  $-\frac{1}{2}$  and the sign convention arise as follows: in the balanced (no-change) state all  $k$  bid components are negative and all  $k$  ask components are positive, so  $\sum_i \text{sign}(X_i) = 0$ . A one-tick upward move converts one ask queue to a bid queue (its sign flips from  $+1$  to  $-1$ ), changing the sum by  $-2$ ; multiplying by  $-\frac{1}{2}$  yields  $\Delta p = +1$  tick, as expected. Symmetrically, a one-tick downward move flips one bid to ask, giving  $\Delta p = -1$  tick.

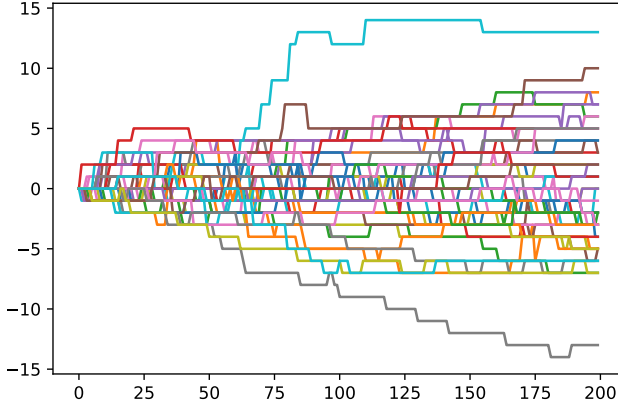


Figure 8: price path real

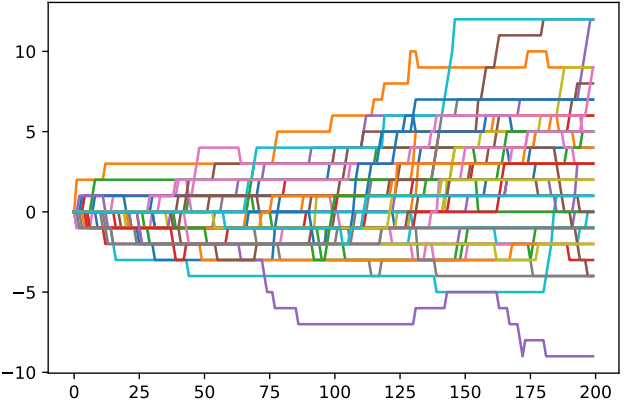


Figure 9: price path model CVAE

As we can see in Figures 8 and 9, price paths are closely matched by the algorithm. Price changes are well respected, giving confidence in the ability to capture the LOB dynamics.

#### 5.5. Reactivity

In that instance, we tested a lift the bid and the join the ask ability. We varied the sizes of the exercise and compared the resulting size-impact relationship with the concave (sub-linear) pattern reported by Ponzi et al. (2009). The setup requires a perfectly balanced lob to maximize the effect and reflect the non arbitrage assumption (Cont et al., 2023). To achieve that, we cloned and inverted our bid model (or ask model). This means that the price path before or after the exercise will be a flat price impact, showcasing only the lift the bid or join the ask operation.

Regarding the model’s reactivity, three tests were performed on our model. The first one aimed at testing the model’s ability to replicate a trajectory observed on the market when their lob information are fed to the simulator on a snapshot by snapshot basis. For this, we identified one period of price decrease and fed it to the model. This setup allows to check the directional ability of the model and prevents, error propagation since forecasts are not stacked on top of each other. That exercise will come later. We also tested the opposite variation by simply inverting the downward trajectory. This exercise allow to assess the reactivity of the model to synthetic data feeding and thus the generalization ability.

Figure 11 shows the various price paths used in the simulation as well as the average path resulting from the liquidity taking action. One interesting fact, visible in Figure 10, is that the effect is more pronounced depending on the amount taken at each 3 seconds. The resulting size-impact relationship is concave: larger interventions produce proportionally smaller additional price displacement, which is qualitatively consistent

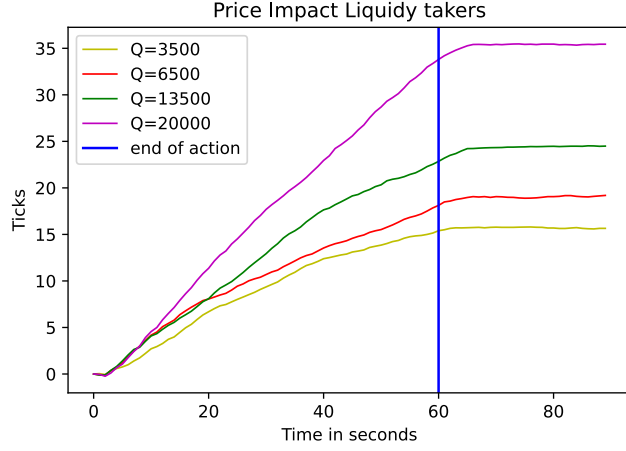


Figure 10: Price Impact liquidity takers: sizes - CVAE

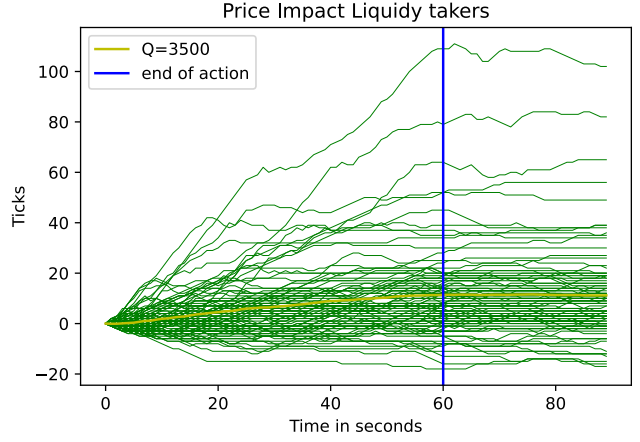


Figure 11: Price Impact liquidity taker - CVAE

with the sub-linear impact reported by Ponzi et al. (2009). However, with only four tested sizes and no formal regression, we do not claim to have identified a specific power-law exponent; a systematic log-log analysis over a finer grid of intervention sizes is left to future work. The price impact persists after the intervention ends (Figures 10 and 12). This is expected under the balanced-book setup: symmetrising the conditioning distribution removes any asymmetry that could drive mean-reversion, so the post-action price remains flat by construction. Characterising transient impact decay would require conditioning on an asymmetric, real-market book state and is left to future work.

Then we did the same exercise but performing a liquidity provision scheme for 60 seconds. The expected effect is a price decrease, as lower prices probability increases.

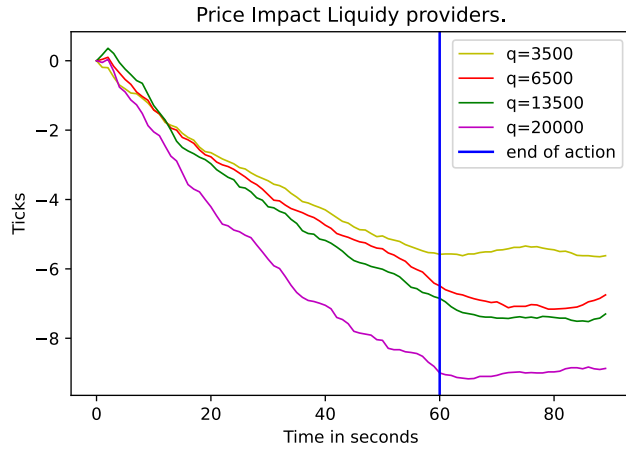


Figure 12: Price Impact liquidity Provider: sizes - CVAE

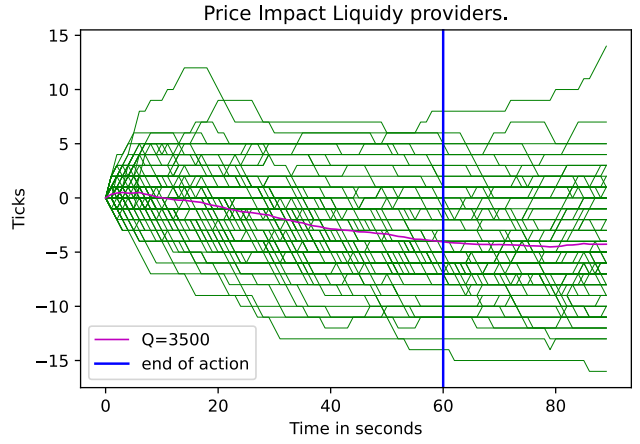


Figure 13: Price Impact liquidity providers - CVAE

Here again, as Figures 12 and 13 show, the size of trades to be added to the best queue affects the price decrease effect. The reactivity is less strong than on the liquidity taking front, which was corroborated in Cont et al. (2023).

## 6. Benchmarking

We benchmark the CVAE against a Conditional Wasserstein GAN with gradient penalty (CGAN) for three reasons. First, conditional adversarial architectures represent the dominant competing deep generative approach in the recent LOB simulation literature: Coletta et al. (2023) and Cont et al. (2023) both use CGAN-style objectives on problems directly comparable to ours, making this the most relevant point of comparison. Second,

the comparison is architecturally controlled: both models receive the same snapshot-based conditioning input  $\mathbf{h}_t$  and produce the same fixed-length depth vector, so differences in performance can be attributed to the training objective rather than to representational capacity or data preprocessing. Third, the tension between likelihood-based objectives (ELBO) and adversarial objectives is a well-documented trade-off in the generative modelling literature (Theis et al., 2016): ELBO training tends to produce smoother but potentially over-smoothed outputs, while adversarial training captures sharper modes at the risk of mode collapse. Evaluating this trade-off empirically on LOB data is the central scientific question of this section.

In order to benchmark our model, we trained a Generative Adversarial Network (GAN)—specifically, a Conditional GAN—on our dataset. GANs are well-recognized generative models composed of two networks: a generator and a discriminator. The generator learns to create high-fidelity outputs (LOB states) from a random input (typically sampled from a Normal distribution). These generated outputs are paired with real LOB states and fed to the discriminator, whose goal is to distinguish genuine outputs from generated ones. A sigmoid activation is applied at the end of the discriminator to produce an output between 0 and 1, which is then used via backpropagation to update the weights of both networks, improving the generator’s ability to fool the discriminator and the discriminator’s ability to detect fakes. As training progresses, the generator learns the characteristics of genuine LOB states and attempts to deceive the discriminator. Ideally, this adversarial game converges to a Nash equilibrium, where both networks reach optimal strategies. However, as argued by Goodfellow et al. (2020), convergence is rarely achieved in practice because optimization occurs in a non-convex function space. When convergence does occur, the discriminator outputs approach 0.5 for all samples, indicating equal likelihood of real or fake. One major advantage of GANs is that they do not require prior distributional assumptions, making them mathematically lighter and highly flexible. However, the two-network adversarial setup introduces significant training challenges. The optimization landscape is a saddle point, which makes training unstable. A well-known issue is mode collapse, where the generator produces limited diversity in outputs. Mode collapse can be full (identical outputs regardless of input noise) or partial (outputs share similar motifs). Durall López et al. (2021) argue that mode collapse is linked to the generator converging toward sharp local minima. When the generator repeatedly produces similar outputs and the discriminator fails to penalize them—often due to vanishing gradients—the generator receives high scores, leading to weak updates and reinforcing collapse. Several techniques mitigate this issue, notably the Wasserstein GAN (W-GAN) and the gradient penalty, which stabilize training by enforcing Lipschitz continuity Arjovsky et al. (2017); Gulrajani et al. (2017). Other challenges include the absence of a formal likelihood or loss metric, complicating feature identification—especially for non-image data like LOB states. Evaluation metrics such as the Fréchet Inception Distance (FID) have been adapted for generative quality assessment ?. To address these issues, we implemented a Conditional W-GAN with gradient penalty, and the following section presents our results.

### 6.1. Evaluation metrics

We compare the two simulators along three axes. Write  $\mathbf{x} = (x_1, \dots, x_{2k})$  for the snapshot vector, with components ordered  $(B_k, \dots, B_1, A_1, \dots, A_k)$  and  $k = 3$ , and let  $P$  and  $\hat{P}$  denote respectively the empirical distribution of the held-out real snapshots and that of an equally sized sample drawn from the trained generator. All three metrics are computed on the same held-out period and are oriented so that lower is better.

The *marginal divergence* aggregates, over the  $2k$  components, the Kullback–Leibler divergence between the one-dimensional empirical marginals  $P_j$  and  $\hat{P}_j$  of component  $j$ . Each per-component divergence is estimated via Gaussian kernel density estimation (KDE) with Silverman bandwidth selection. Both densities are evaluated on a shared grid of  $n = 1000$  equally spaced points covering the union of their supports, re-normalised to integrate to unity, and stabilised by adding  $\varepsilon = 10^{-10}$ . The divergence is then computed as

$$D_{\text{KL}}(P_j \| \hat{P}_j) = \int p_j(x) \ln \frac{p_j(x)}{\hat{p}_j(x)} dx \approx \sum_{i=1}^n p_j(x_i) \ln \frac{p_j(x_i)}{\hat{p}_j(x_i)} \Delta x,$$

and the reported metric averages over the  $2k$  components:

$$D_{\text{KL}}^{\text{marg}} = \frac{1}{2k} \sum_{j=1}^{2k} D_{\text{KL}}(P_j \| \hat{P}_j).$$

The *queue error* measures the discrepancy between the average depths displayed at each level, aggregated

over the  $2k$  components,

$$E_{\text{queue}} = \frac{1}{2k} \sum_{j=1}^{2k} \left| \mathbb{E}_P[x_j] - \mathbb{E}_{\hat{P}}[x_j] \right|.$$

The *correlation error* compares the full level-to-level dependence structure through the Frobenius norm of the difference between the two  $2k \times 2k$  Pearson correlation matrices  $C$  and  $\hat{C}$ ,

$$E_{\text{corr}} = \|C - \hat{C}\|_F = \left( \sum_{i=1}^{2k} \sum_{j=1}^{2k} (C_{ij} - \hat{C}_{ij})^2 \right)^{1/2}.$$

### 6.2. Marginal distribution

The three marginal distributions (Figures 14, 15, and 16) illustrate that the CGAN manages to reproduce the empirical shapes with high fidelity across all liquidity levels. Compared to the CVAE, the CGAN tends to capture sharper local modes, which is consistent with the adversarial training objective that pushes generated samples toward realistic high-density regions. However, this behaviour sometimes comes at the cost of slightly higher variance in low-density tails, a pattern commonly observed in GAN-based simulators.

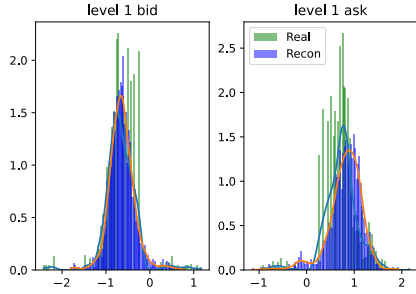


Figure 14: Level 1 GAN

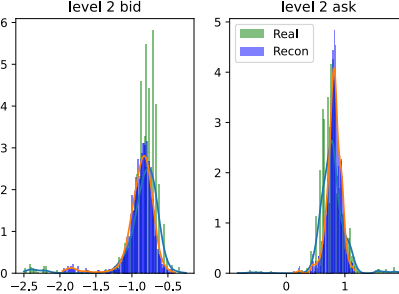


Figure 15: Level 2 GAN

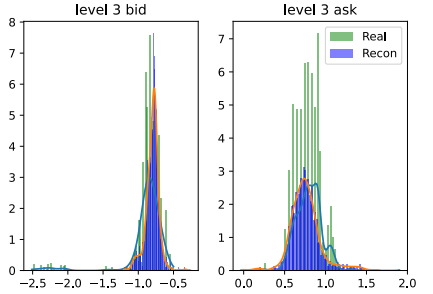


Figure 16: Level 3 GAN

This is an important distinction between the CGAN and CVAE frameworks. While the CVAE explicitly learns a conditional likelihood, the CGAN does not assume nor impose any parametric form on the marginals. This allows the CGAN to replicate complex empirical shapes naturally, especially when the data exhibits multimodality or skewness. The absence of likelihood-based optimization, however, also means that training stability must be carefully monitored to avoid mode collapse (Goodfellow et al., 2020; Durall López et al., 2021). Crucially, sharper marginals do not automatically translate into better structural fidelity: as the correlation and queue-level results below show, the CVAE’s likelihood-based objective produces superior joint dependence structure despite its higher marginal KL, suggesting that for LOB simulation—where inter-level coherence and queue accuracy are operationally more important than marginal sharpness—the CVAE’s training objective is better aligned with the evaluation criteria. A multimodal CVAE could help bridge the gap between the two approaches.

### 6.3. Average Best Quantities

The average best quantities produced by the CGAN (Figure 17, as described in Section 5.2) closely aligns with the empirical benchmark, indicating that the model captures not only marginal behavior but also the relative ordering of optimal liquidity placement across levels. Overall, the agreement between the CGAN-generated averages and the real data further supports the model’s ability to learn meaningful liquidity allocation patterns. Both models reproduce the LOB smile, but the CVAE achieves a lower queue-level error (34.10 vs. 55.30 shares; see Table 3), indicating that its likelihood-based objective produces more accurate queue size estimates—a practically significant advantage for execution cost analysis and market-making strategy evaluation.

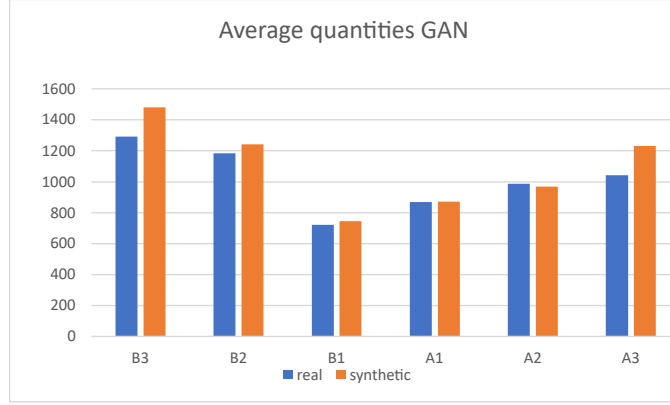


Figure 17: average best quantities

#### 6.4. Level-to-level correlation

Correlations between liquidity levels are a key structural element of the simulator. The CGAN maintains these dependencies reasonably well (as exhibited in Figures 18 and 19), especially for adjacent levels where empirical co-movements are strongest. However, the CVAE achieves a lower Frobenius norm between the empirical and generated correlation matrices (0.34 vs. 0.44), indicating that it preserves the full inter-level dependence structure more faithfully. This is consistent with the observation that the ELBO objective, by penalising reconstruction errors at each level simultaneously, implicitly regularises the joint distribution of queue sizes across levels. The adversarial objective, by contrast, evaluates entire joint samples holistically, which can sharpen individual modes while introducing artefacts in cross-level relationships.

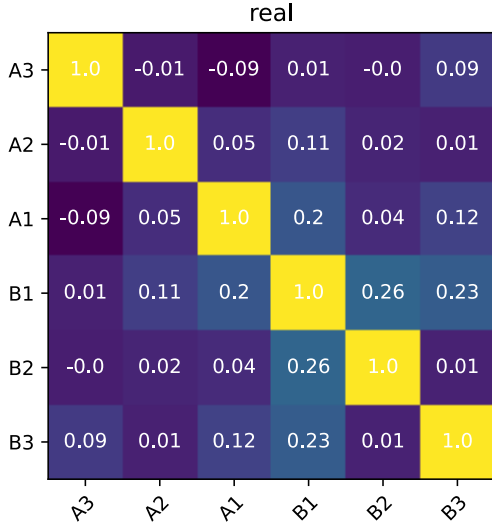


Figure 18: Correlations realGAN



Figure 19: Correlation model generated GAN

#### 6.5. Reactivity

The GAN also shows a liquidity taking reactivity (Figure 20), which is line with literature and brings comfort in the CVAE results.

The reactivity plot highlights that the CGAN effectively captures the liquidity taking dynamics observed in the empirical data. In particular, the average path responds in the expected direction as depth is consumed at the best quote, demonstrating that aggressive order flow in the simulation moves the price consistently with the structural patterns identified in real markets. This ability to produce a directional price response confirms that the CGAN does not simply memorize marginal distributions but learns a meaningful liquidity–response

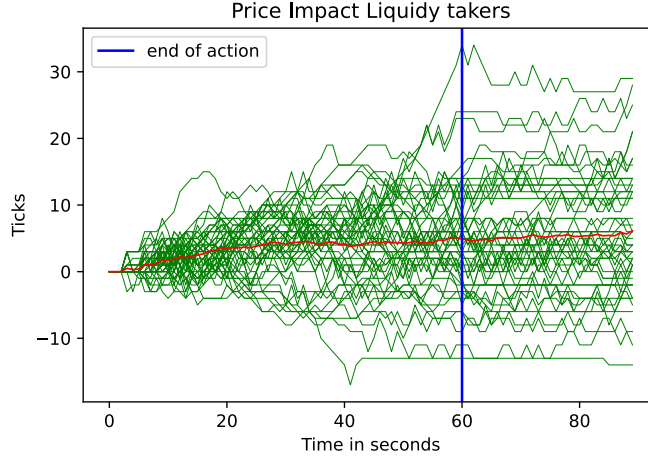


Figure 20: Liquidity takers GAN

mechanism. The resulting behaviour is coherent with established findings in the microstructure literature, thereby reinforcing the reliability of the CVAE’s own reactivity results by providing an independently trained adversarial benchmark.

We also provided a quantitative comparison table between the CVAE and the CGAN approach.

| Stock | Model | Marginals KL divergence | Queue error (shares) | Correlation Diff Frobenius norm |
|-------|-------|-------------------------|----------------------|---------------------------------|
| BMO   | CVAE  | 0.1434                  | 34.0966              | 0.3389                          |
| BMO   | CGAN  | 0.0542                  | 55.2965              | 0.4371                          |
| Telus | CVAE  | 0.0601                  | 52.7655              | 0.2500                          |
| Telus | CGAN  | 0.0662                  | 162.4581             | 0.7055                          |

Table 3: Performance comparison of CVAE and CGAN

The CGAN achieves a lower marginals KL divergence on BMO (0.054 vs. 0.143), reflecting its ability to produce sharper local modes consistent with the adversarial objective’s emphasis on sample realism. On Telus, however, the marginal advantage reverses (0.066 vs. 0.060), suggesting that on less liquid stocks the CVAE’s likelihood-based objective is already competitive on marginal fidelity. The CVAE consistently outperforms on the two metrics most directly relevant to LOB simulation practice across both stocks: queue-level fidelity (BMO: 34.10 vs. 55.30, a 38% improvement; Telus: 52.77 vs. 162.46, a 68% improvement) and inter-level correlation structure (BMO: 0.34 vs. 0.44; Telus: 0.25 vs. 0.71). Notably, the CGAN’s structural degradation is more severe on the less liquid Telus stock, where the adversarial objective’s difficulty in capturing thin-queue dynamics leads to a nearly threefold increase in queue error relative to the CVAE. Accurate queue sizes directly affect execution cost estimates, and correct correlation structure is essential for multi-level order placement strategies. The CVAE requires no gradient-penalty coefficient or critic-iteration schedule and is implemented as a single network—substantially reducing the hyperparameter search burden. Taken together, the quantitative results across both stocks support the CVAE as the preferred model when structural fidelity and training simplicity are prioritised over marginal sharpness, which is the relevant criterion for production-level LOB simulators.

## 7. Ablation Study

To quantify the contribution of the LSTM history encoder, we perform an ablation study on the conditioning sequence length  $h$ . We train the CVAE under identical hyperparameters for  $h \in \{1, 2, 4, 10, 20\}$ , where  $h = 1$  corresponds to a first-order Markov model that conditions only on the immediately preceding snapshot. Each configuration is trained with 5 independent random seeds; the test-set mean squared error (MSE) is reported in Table 4 as mean  $\pm$  one standard deviation.

| Lag $h$ | Telus                                 | BMO                                   |
|---------|---------------------------------------|---------------------------------------|
| 1       | $0.0259 \pm 0.0029$                   | $0.0396 \pm 0.0068$                   |
| 2       | $0.0219 \pm 0.0013$                   | $0.0370 \pm 0.0024$                   |
| 4       | $0.0192 \pm 0.0005$                   | $0.0360 \pm 0.0019$                   |
| 10      | $0.0181 \pm 0.0011$                   | $0.0344 \pm 0.0018$                   |
| 20      | <b><math>0.0158 \pm 0.0010</math></b> | <b><math>0.0312 \pm 0.0012</math></b> |

Table 4: Test MSE as a function of LSTM conditioning length  $h$  (mean  $\pm$  std over 5 seeds).

Both stocks exhibit a monotonic and statistically significant improvement as the conditioning horizon increases. For BMO, MSE decreases from  $0.0396 \pm 0.0068$  at  $h = 1$  to  $0.0312 \pm 0.0012$  at  $h = 20$  (a 21% reduction), with non-overlapping confidence intervals between extreme configurations. For Telus, MSE decreases from  $0.0259 \pm 0.0029$  at  $h = 1$  to  $0.0158 \pm 0.0010$  at  $h = 20$  (a 39% reduction), confirming that the benefit of longer temporal context holds across liquidity regimes. The monotonic trend is consistent with the intuition that longer context helps the LSTM track queue replenishment dynamics, while the larger relative gain on Telus suggests that the sparser informative transitions in less liquid books benefit more from extended memory.

We retain  $h = 10$  for the main experiments as it provides most of the improvement (30% on Telus, 13% on BMO relative to  $h = 1$ ) at moderate computational cost, noting that  $h = 20$  offers further gains at the expense of longer conditioning sequences.

## 8. Conclusion

In this paper, we have presented a reactive limit order book simulator combining a Conditional Variational Autoencoder with an LSTM history encoder, trained and evaluated on Toronto Stock Exchange equity data. The simulator reproduces the four key stylized facts examined — marginal queue distributions, the average-quantity smile, level-to-level correlations, and realistic price paths — and exhibits reactive price impact for both liquidity taking and provision, with a concave size-impact relationship qualitatively consistent with the literature. Benchmarked against a Conditional WGAN-GP trained on the same data, the CVAE achieves superior queue-level fidelity (34.10 vs. 55.30 shares) and better inter-level correlation structure (Frobenius norm 0.34 vs. 0.44), at the cost of higher marginal KL divergence (0.143 vs. 0.054). The CVAE’s single-network ELBO objective avoids the gradient-penalty tuning and critic-iteration scheduling required by the WGAN-GP, substantially reducing the hyperparameter search burden. These properties make the CVAE a competitive and practical alternative to adversarial approaches for production-level LOB simulation, particularly when structural fidelity and training simplicity are prioritised over marginal sharpness.

A limitation of this study is the age of the dataset: while the architecture and training procedure are not specific to that period, market microstructure evolves over time; for instance through changes in tick-size regimes, fee schedules, or the prevalence of algorithmic participants. Validation on more recent data would strengthen the generalisability claim.

Several promising avenues can be pursued to further extend and strengthen the proposed framework. First, the data encoding mechanism could be refined to explicitly handle non-zero bid–ask spreads and discontinuities in level prices, while preserving the overall structure of the model and its homogeneity property. Such an extension would significantly broaden the applicability of the approach, enabling its use on less liquid equities as well as other asset classes, including foreign exchange, interest rates, and commodities. Second, departures from the Gaussian prior assumption could be investigated through a multimodal prior framework. Empirical evidence suggests that marginal distributions often exhibit multimodality, which may not be adequately captured by the current isotropic Gaussian prior. This limitation could be addressed by adopting more expressive prior distributions, such as mixtures of Gaussians or normalising flows, thereby improving the model’s ability to represent complex distributional features. Finally, a foundation model approach could be explored to support joint training and inference across multiple stocks. The relatively simple and scalable training scheme facilitates the use of longer time horizons and richer datasets, opening the door to cross-asset learning and improved generalisation through shared representations (Ansari et al.; Das et al., 2024).

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